Retrieval-Augmented Generation

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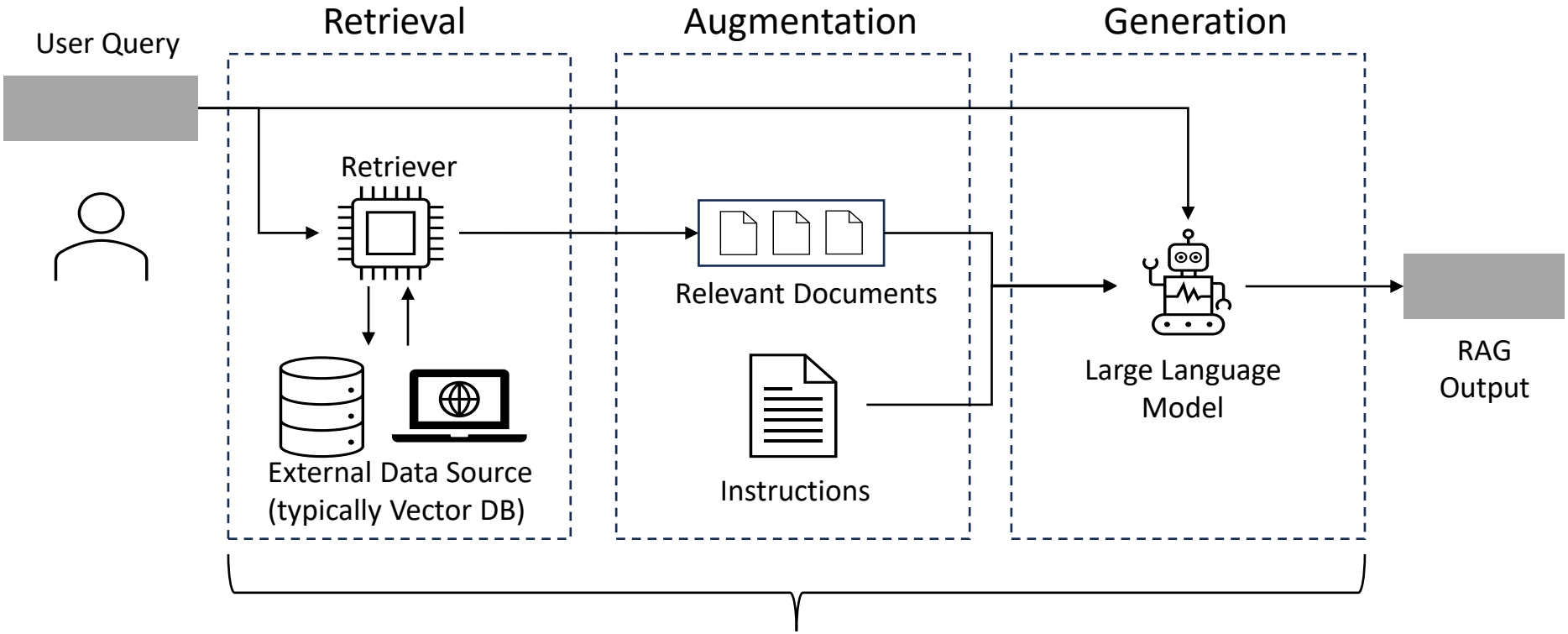
Retrieval-Augmented Generation:

General Workflow

Retrieval-Augmented Generation

General Workflow



Retrieval-Augmented Generation Pipeline

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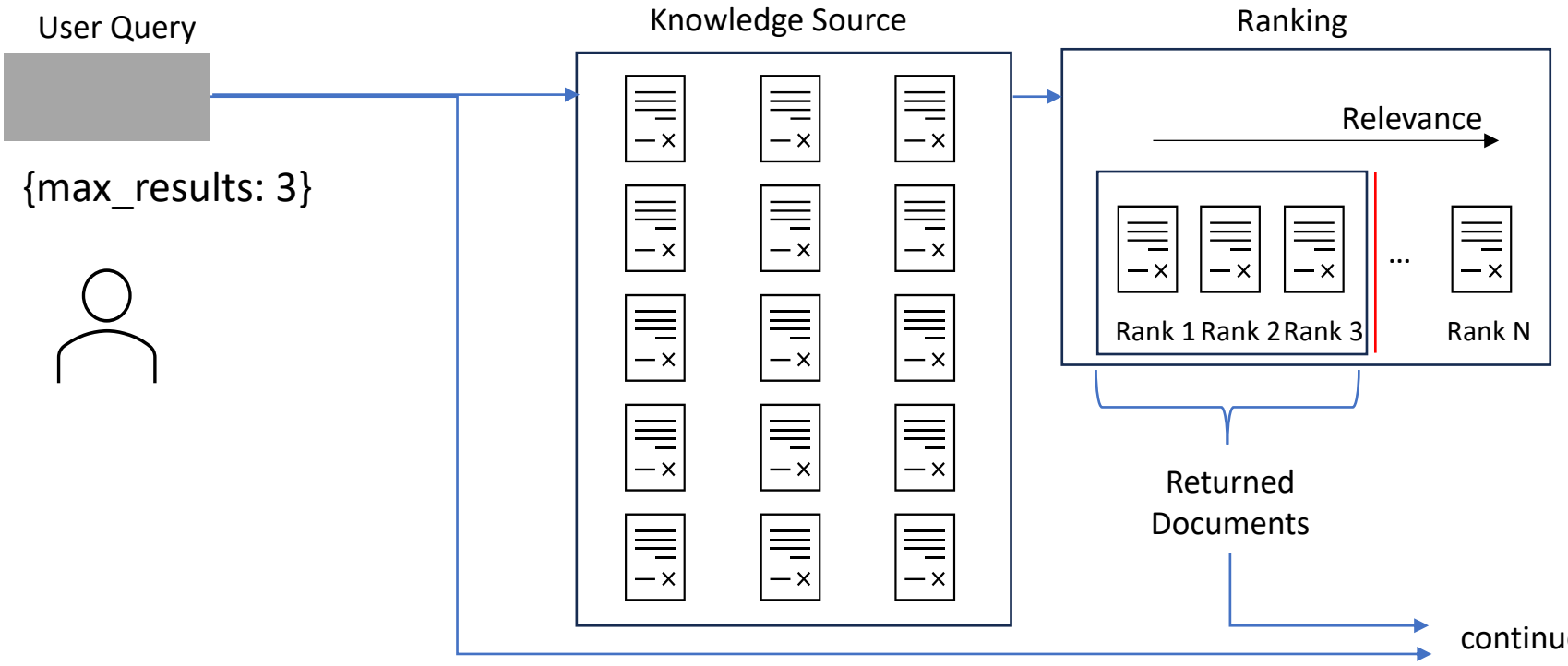
Retrieval-Augmented Generation

Retrieval Process



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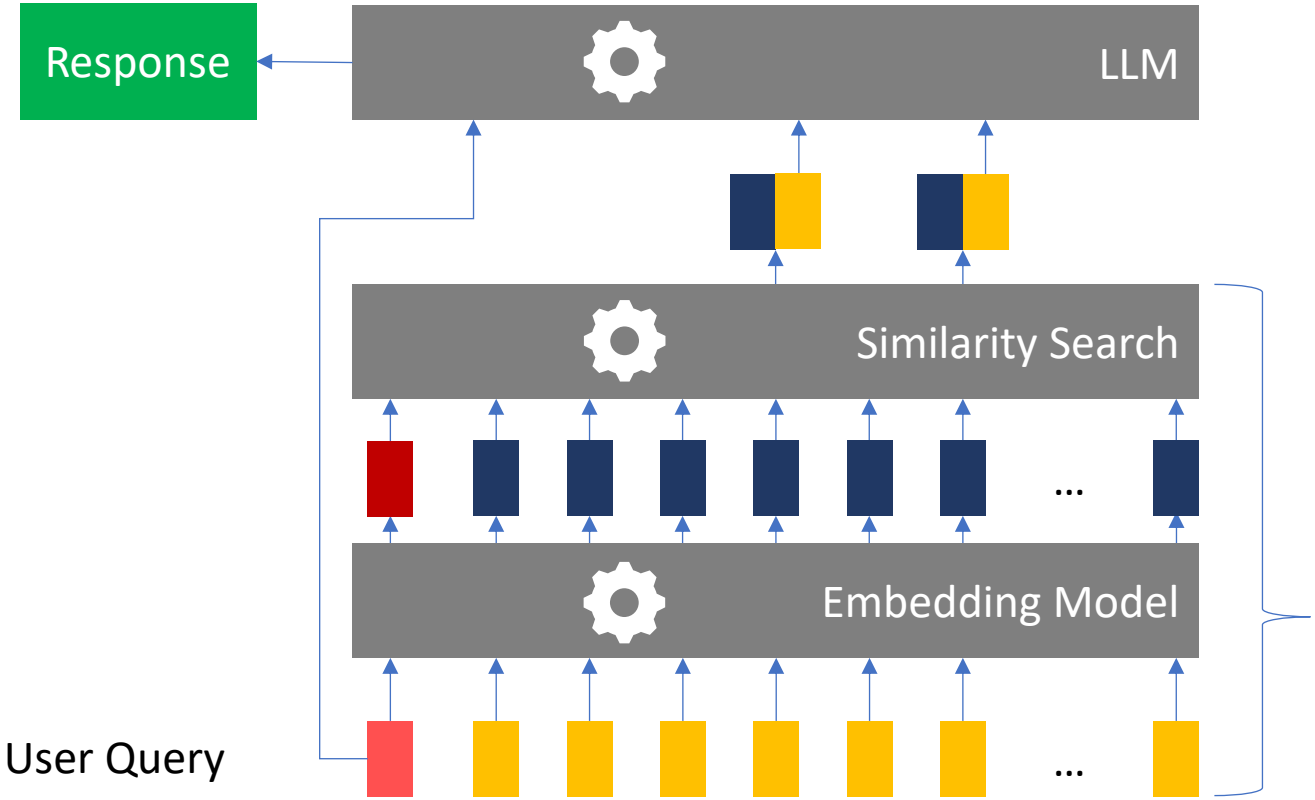
Retrieval-Augmented Generation:

Simple RAG

Retrieval-Augmented Generation

Workflow with Vector DB Backend



Nearest Neighbors

Embeddings

Vector Database

Text Documents

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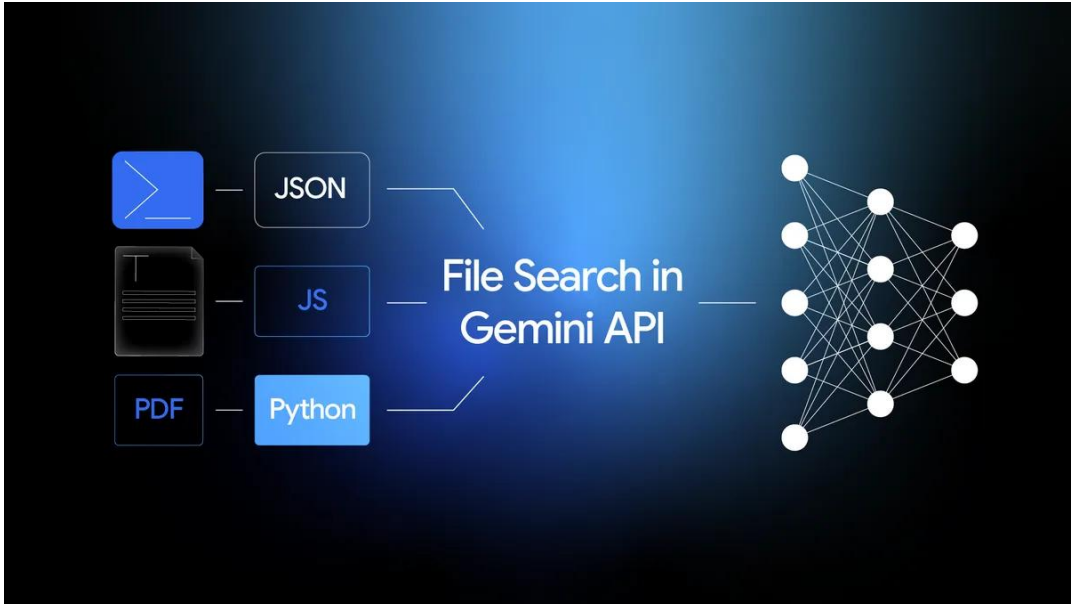
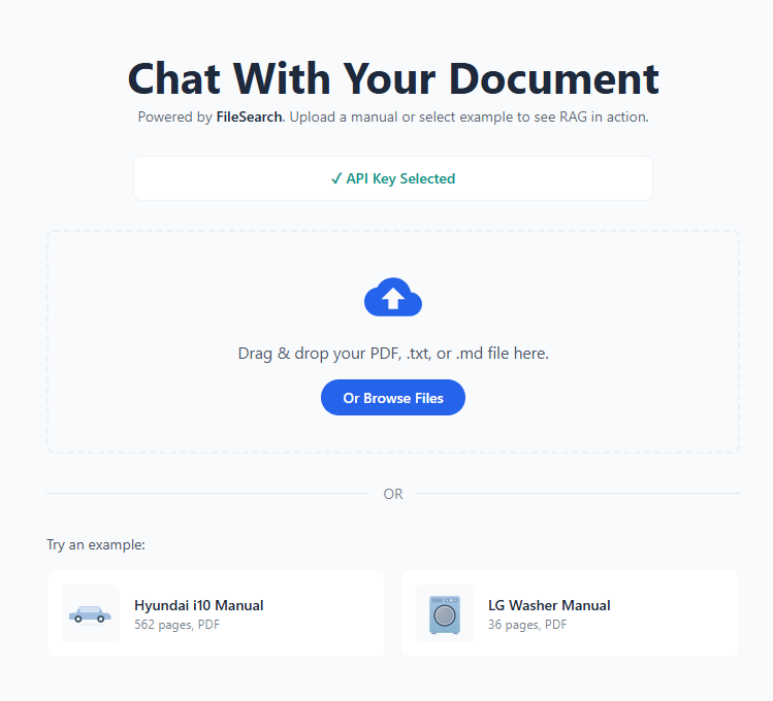
Retrieval-Augmented Generation

Google File Search

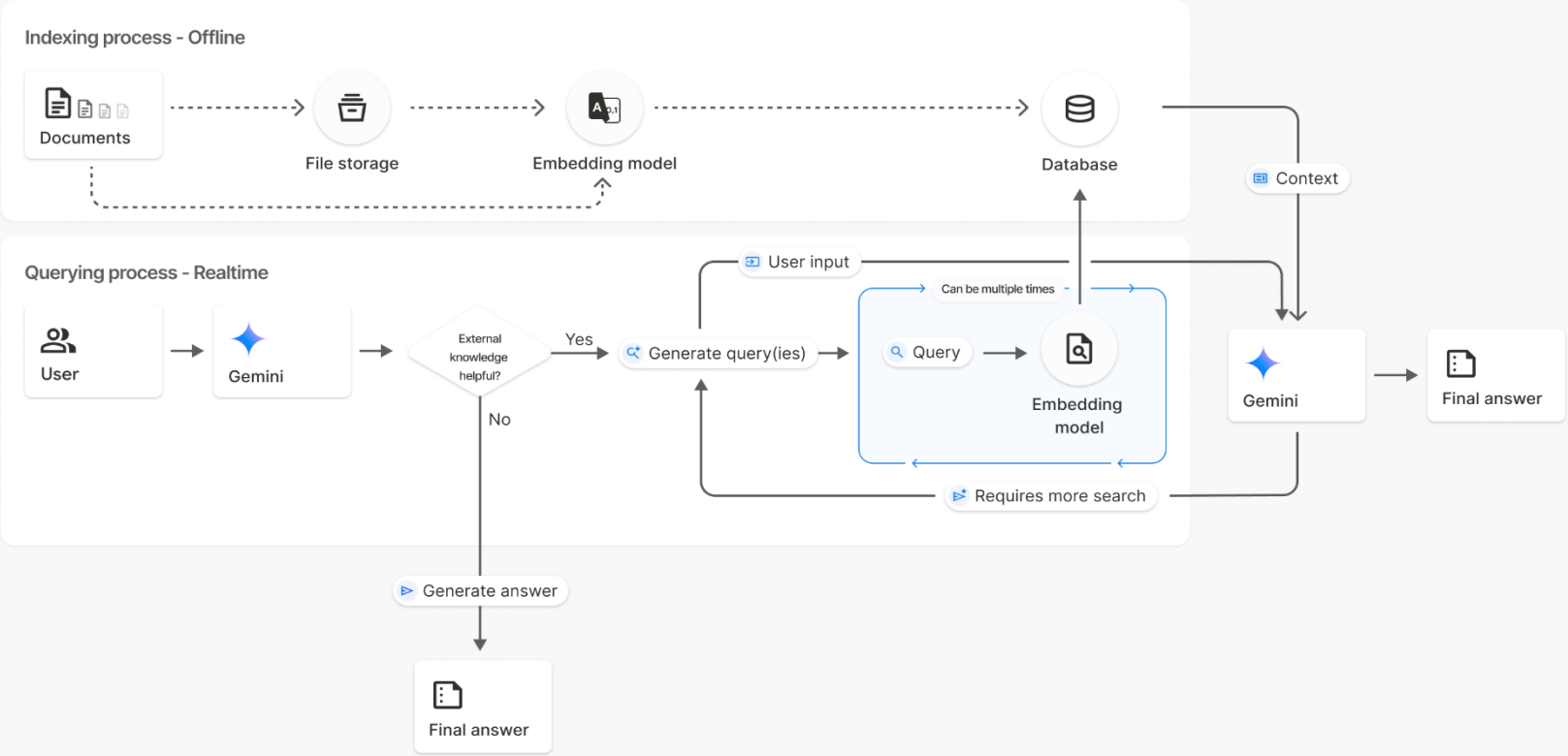




Source: https://aistudio.google.com/apps/bundled/ask_the_manual?showPreview=true&showAssistant=true

Retrieval-Augmented Generation

Google File Search



Retrieval-Augmented Generation:

RAG Improvements for Pre-Retrieval

Retrieval-Augmented Generation

Levels of RAG

Explicit Fact Queries

Level 1

- known facts
- no reasoning

What is the majority party in Germany?

Implicit Fact Queries

Level 2

- known facts
- interpretation needed to identify answer

What is the majority party in the country where Berlin is located?

Interpretable Rationale Queries

Level 3

- understand reasoning
- logical explanation

Hidden Rationale Queries

Level 4

- deeper insights
- requires context-based reasoning

Source: https://arxiv.org/pdf/2409.14924

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Retrieval-Augmented Generation

RAG Improvement for Pre-Retrieval, Retrieval, and Post-Retrieval

Pre-Retrieval

- improve indexing pipeline

Retrieval

- improve retrieval process

Post-Retrieval

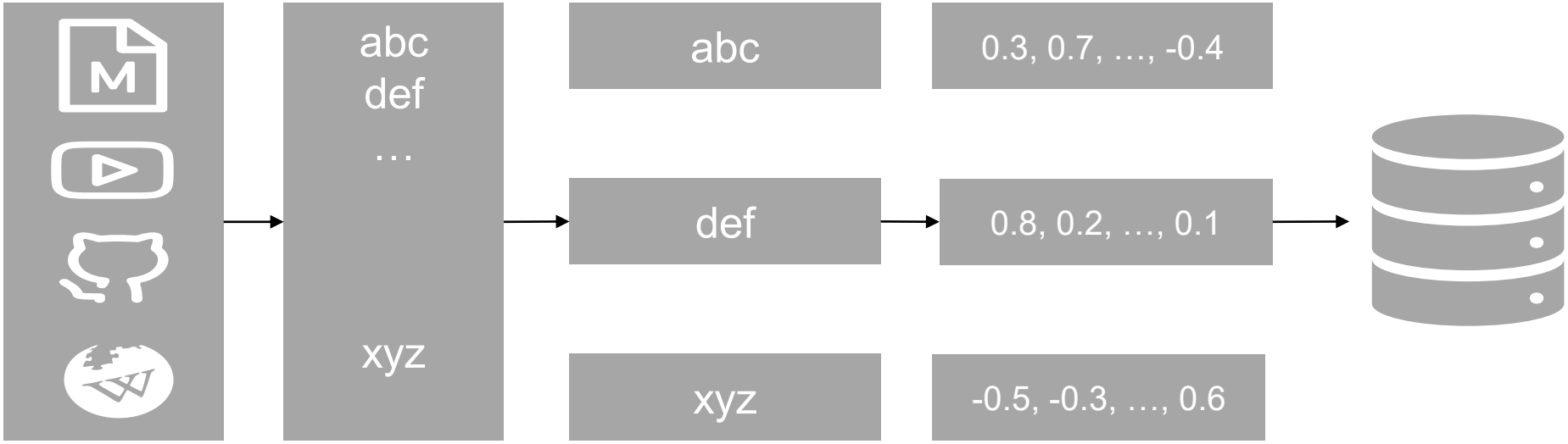
- improve user query

Retrieval-Augmented Generation:

RAG Improvements for Pre-Retrieval

Retrieval-Augmented Generation

RAG Improvement Approaches Pre-Retrieval



Data Ingestion Optimization Techniques

Metadata Enhancing	Chunk Size Optimization	Embedding Model Finetuning
Data Cleaning	Context Enrichment	

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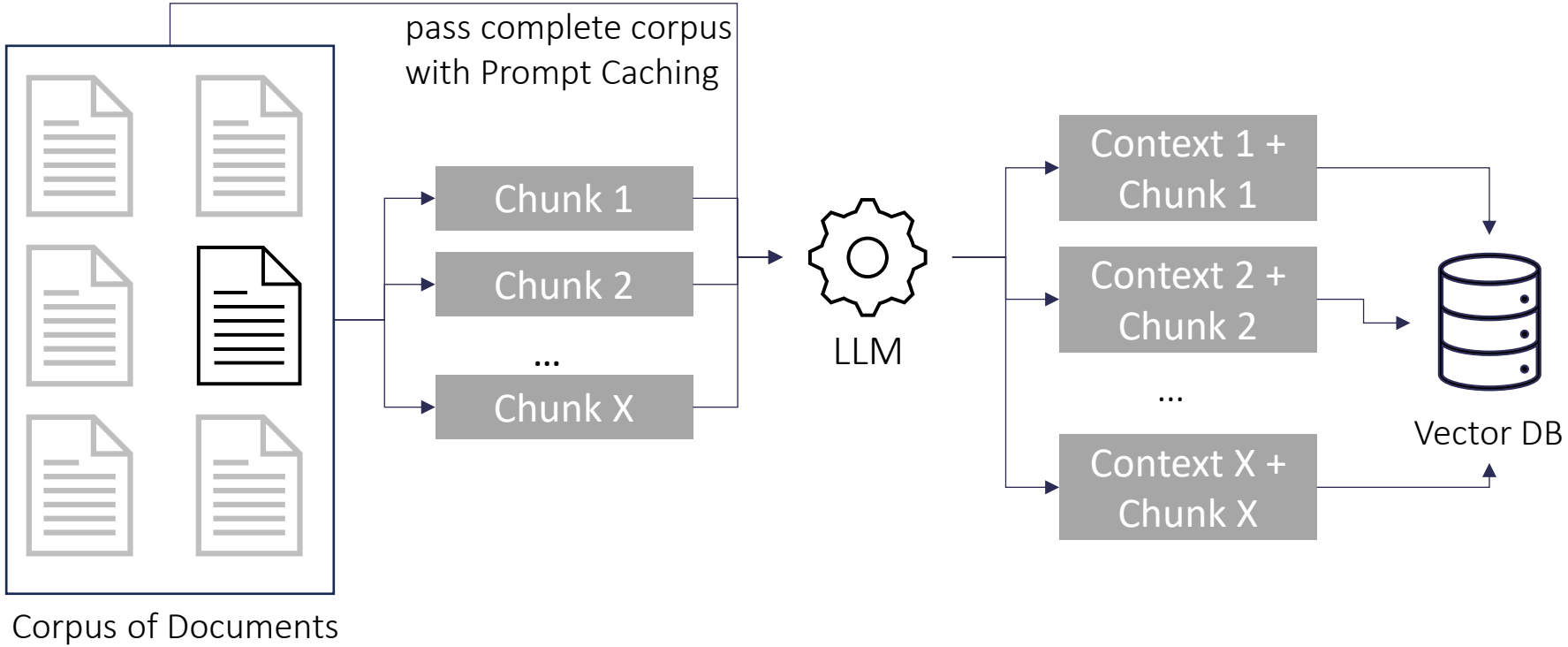
Retrieval-Augmented Generation

RAG Improvement for Pre-Retrieval: Context Enrichment with Contextual Retriever



own graph; adapted from https://www.anthropic.com/news/contextual-retrieval

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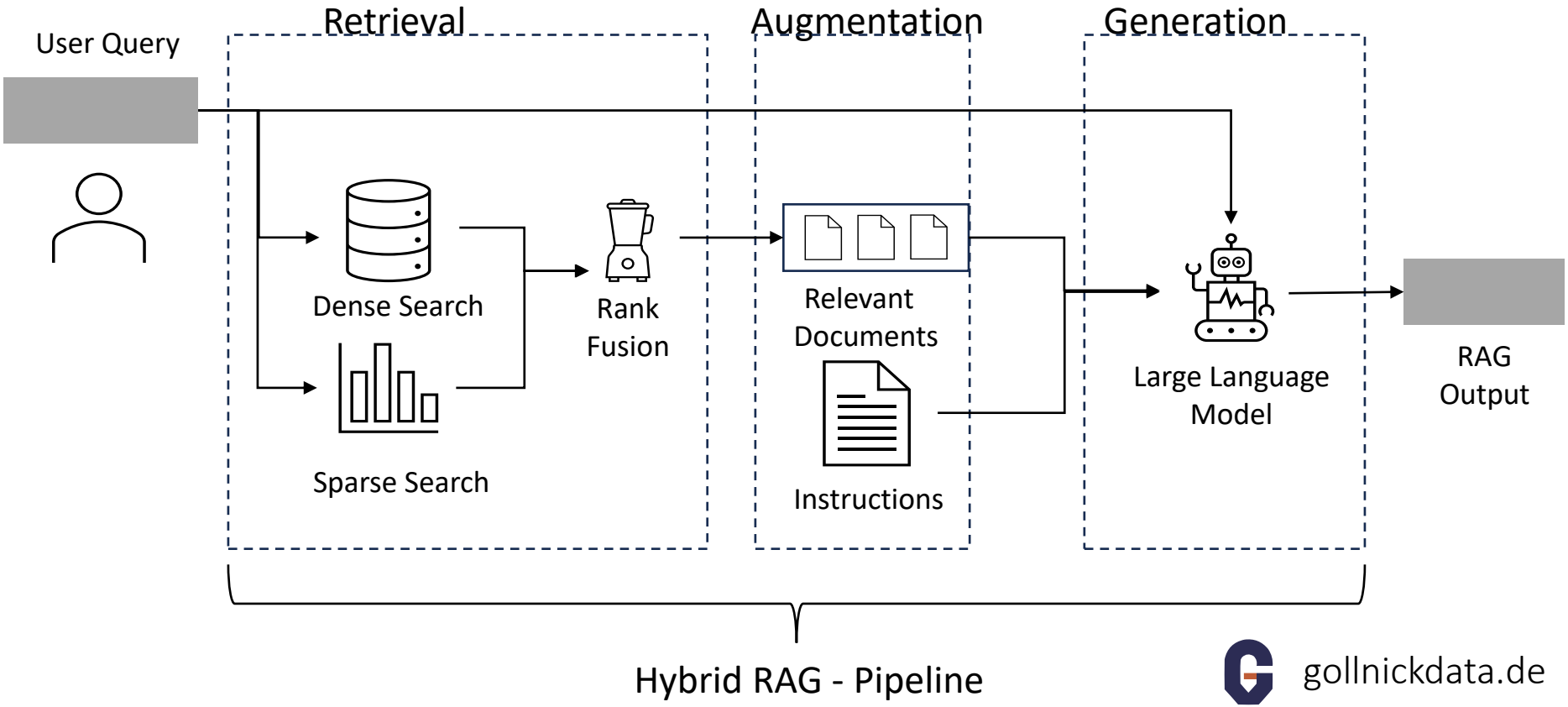
Retrieval-Augmented Generation:

RAG Improvements for Retrieval

Retrieval-Augmented Generation

RAG Improvement for Retrieval: Hybrid RAG



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Retrieval-Augmented Generation

RAG Improvement for Retrieval: Hybrid RAG

🔍 Sparse vs Dense Vector Search

Diese App verdeutlicht den Unterschied zwischen **Sparse Search** (Keyword-basiert) und **Dense Search** (Semantik-basiert). Geben Sie Texte ein, erstellen Sie den Index und vergleichen Sie die Suchergebnisse.

1. Dokumente hinzufügen

Geben Sie hier Ihre Dokumente ein (eine Zeile pro Dokument):

```
Generative KI verändert die Art wie wir arbeiten.
BG Ein Hund spielt im Park mit einem Ball.
Die Katze liegt faul auf der Matte.
Künstliche Intelligenz und Machine Learning sind spannende Themen.
BG Feline Vierbeiner genießen die Sonne.
Deep Learning ist ein Teilbereich von AI.
Ein Haustier kann viel Freude bereiten.
```

Index erstellen

2. Suchen & Vergleichen

Geben Sie einen Suchbegriff ein:

BG

Sparse Search (BM25)

Basiert auf exakten Wortübereinstimmungen und Worthäufigkeit.

	Dokument	Score
4	BG Feline Vierbeiner genießen die Sonne.	0.8565
1	BG Ein Hund spielt im Park mit einem Ball.	0.7130

🧠 Dense Search (Embeddings)

Basiert auf der semantischen Bedeutung (Vektoren).

	Dokument	Abstand (Score)
0	BG Feline Vierbeiner genießen die Sonne.	0.3596
1	BG Ein Hund spielt im Park mit einem Ball.	0.3666

Retrieval-Augmented Generation

Reciprocal Rank Fusion

Rank	Dense Search Result	Sparse Search Result	Reciprocal Rank
1	Document D	Document A	1/1
2	Document A	Document C	1/2
3	Document C	Document B	1/3
4	Document B	Document D	1/4

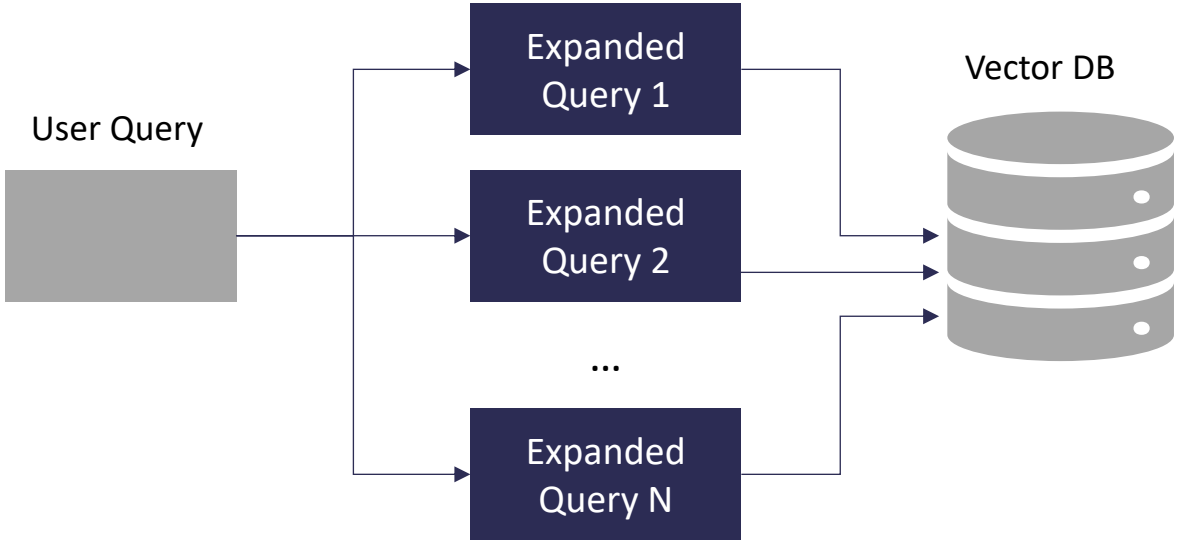
a) Reciprocal Rank Calculation

Document A	1/2 + 1/1 = 1.5
Document D	1/1 + 1/4 = 1.25
Document C	1/3 + 1/2 = 0.83
Document B	1/4 + 1/3 = 0.58

b) Reciprocal Rank Fusion

Retrieval-Augmented Generation

RAG Improvement for Retrieval: Query Expansion



Retrieval-Augmented Generation

RAG Improvement for Retrieval: Query Expansion

Synonym Expansion

expand with synonyms
“Climate change” →
[“global warming”,
“climate change”,
“environmental impact”]

Related Terms Expansion

related terms
“Machine learning” →
[“machine learning”,
“deep learning”,
“artificial intelligence”, “neural networks”]

Conceptual Expansion

“Renewable energy” →
[“renewable energy”,
“solar power”,
“wind energy”,
“green technology”,
“sustainable energy”]

Retrieval-Augmented Generation

RAG Improvement for Retrieval: Query Expansion

Phrase Variation Expansion

including variations of how a concept might be phrased ensures retrieval from documents that use different wording to discuss the same idea.
“Health benefits of exercise” →
[“health benefits of physical activity”,
“exercise health benefits”,
“positive effects of exercise”]

Contextual Expansion

Including related fields or tasks
“Natural language processing” →
[“natural language processing”,
“text analysis”,
“computational linguistics”,
“language modeling”]

Temporal Expansion

Expanding for common abbreviations and alternative names →
[“COVID-19 vaccines”,
“COVID vaccines”,
“coronavirus immunization”,
“SARS-CoV-2 vaccine”]

Entity-Based Expansion

When query involves an entity, expanding with related people, products, or attributes associated with the entity can enhance retrieval quality.
“Tesla” →
[“Tesla”, “Elon Musk”,
“electrical vehicles”,
“autonomous driving”]

Retrieval-Augmented Generation:

RAG Improvements for Post-Retrieval

Retrieval-Augmented Generation

RAG Improvement for Post-Retrieval: Prompt Compression

User Query

very long context

- long context
- increased noise
- reduced LLM performance

Small LLM

- cheap LLM

Compressed Query

irrelevant tokens removed

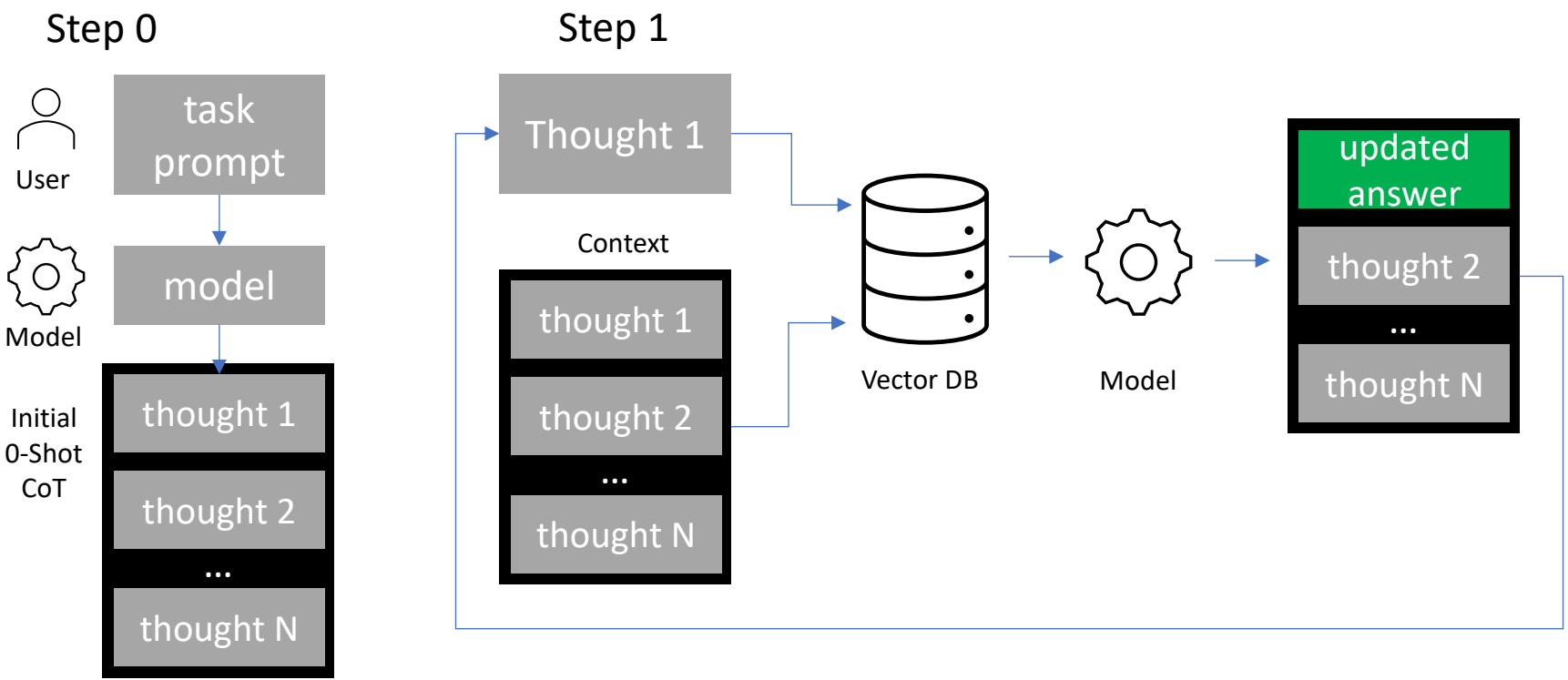
- irrelevant tokens removed

LLM

- powerful LLM
- cost ↓
- performance ↑
- efficiency ↑

Retrieval-Augmented Generation

Retrieval Augmented Thought



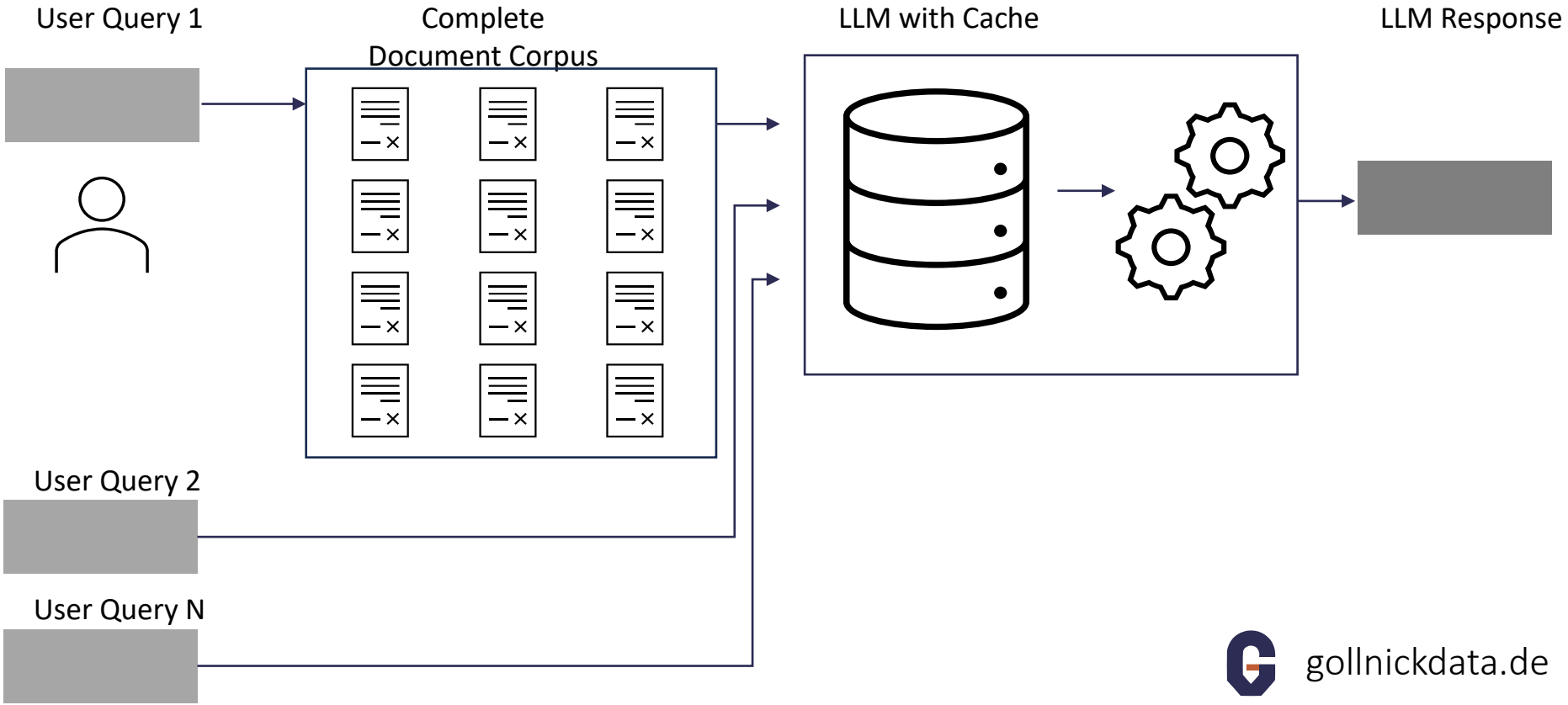
Source: Zihao Wang, et. al. „RAT: Retrieval Augmented Thoughts Elicit Context-Aware Reasoning in Long-Horizon Generation“

Retrieval-Augmented Generation:

RAG Alternatives

Retrieval-Augmented Generation

Prompt Caching



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